

HEALTHCARE DATA CLIENT

Data Validation & Automation Testing

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Test Scope	Data validation, exception handling, reconciliation and Power BI reporting
Processing Scale	Approximately 250,000 records per batch
Confirmed Result Context	2% exception rate; review turnaround reduced from 6 hours to 2.5 hours per file

Testing status: Testing formed part of the engagement, but exact historical test-case IDs, execution logs, defect counts, formal UAT records and sign-off are not retained. The test cases in this document are a portfolio reconstruction aligned to confirmed requirements and rules.

1. Testing Objective

Verify that the automated validation workflow identifies applicable data-quality failures correctly, produces traceable exception outputs, preserves record-level KPI integrity and reconciles Power BI reporting to governed validation results.

2. Test Scope

In Scope	Not Claimed
Source population reconciliation	Original client test scripts
Eight confirmed validation categories	Historical defect counts
Exception flagging and reason assignment	Formal historical UAT sign-off
Record-level versus rule-level grain	Unretained clinical validation logic
Exception Rate calculation	Unretained healthcare thresholds or code sets
Batch-scale processing reference	Specific production performance SLA
Power BI reconciliation	Exact production refresh architecture

3. Test Approach

Test Layer	Purpose
Data intake testing	Confirm supported inputs enter the governed validation process and record populations can be reconciled.
Rule testing	Verify each implemented validation rule produces the expected pass or fail result.
Exception testing	Confirm failed records are flagged, classified and routed to the exception output.
Grain testing	Verify multiple failures on one record do not inflate record-level exception metrics.
Reconciliation testing	Validate processed, exception and rule-failure populations against each other.
Reporting testing	Confirm Power BI measures reconcile to the validation output.
Regression testing	Retest affected logic when rule parameters or reporting logic change.

4. Core Validation Test Cases

Test ID	Area	Test Condition	Expected Result	Trace
TC-DQ-01	Duplicate detection	Input includes records satisfying and not satisfying the implemented duplicate definition.	Only records meeting the documented duplicate rule are flagged.	DQ-01; US-02
TC-DQ-02	Missing values	Applicable field contains populated, null and blank examples.	Missing values fail only where the completeness rule applies.	DQ-02; US-02
TC-DQ-03	Invalid dates	Applicable date field contains valid and invalid examples under the implemented date rule.	Invalid values are flagged; no replacement date is fabricated.	DQ-03; US-03
TC-DQ-04	Incorrect formats	Values include conforming and non-conforming examples.	Only values failing the documented format rule are flagged.	DQ-04; US-03
TC-DQ-05	Invalid identifiers	Identifier values include valid and invalid examples under the governed ID rule.	Only identifiers failing the documented rule are flagged.	DQ-05; US-03
TC-DQ-06	Range validation	Applicable values include in-range, boundary and out-of-range examples.	Values outside documented bounds are flagged correctly.	DQ-06; US-03
TC-DQ-07	Code consistency	Values include governed and non-governed code examples.	Values inconsistent with the documented code set or mapping are flagged.	DQ-07; US-04
TC-DQ-08	Cross-field mismatch	Related fields include records satisfying and violating the documented relationship.	Only documented relationship violations are flagged.	DQ-08; US-04

5. Exception Handling Test Cases

Test ID	Test Condition	Expected Result	Trace
TC-EX-01	Record fails one applicable rule.	Record is flagged as one Exception Record and has an identifiable failure reason.	US-05, US-06
TC-EX-02	Record fails multiple applicable rules.	Record remains one Exception Record and retains multiple rule-level failures.	US-05, US-06
TC-EX-03	Record passes all applicable rules.	Record is not included in the exception population.	US-05
TC-EX-04	Exception output is generated.	Flagged records are available in a separate file, table or report.	BR-11
TC-EX-05	Reason information is inspected.	Each failure has a reason code or equivalent explanation.	BR-12
TC-EX-06	Source reconciliation is performed.	Exception can be traced to the affected processed source record.	BR-17

6. KPI & Grain Test Cases

Test ID	Control	Expected Result
TC-KPI-01	Records Processed	Matches the governed processed batch population.
TC-KPI-02	Exception Records	Counts distinct records failing one or more applicable rules.
TC-KPI-03	Exception Rate	Exception Records / Records Processed x 100.
TC-KPI-04	Multi-failure record	Multiple rule failures do not increase the record-level exception count beyond one for that record.
TC-KPI-05	Validation Failures	Rule-level count can exceed Exception Records when records fail multiple rules.
TC-KPI-06	Observed result	Retained engagement Exception Rate is 2%, not treated as a target or SLA.
TC-KPI-07	Turnaround calculation	6.0 hours minus 2.5 hours equals 3.5 hours reduction.
TC-KPI-08	Reduction percentage	3.5 / 6.0 x 100 equals approximately 58.3% turnaround reduction.

7. Workflow Test Cases

Test ID	Workflow Stage	Expected Control
TC-WF-01	Ingest	Supported Excel, CSV or SQL/database input can enter the process.
TC-WF-02	Validate	Applicable documented rules are executed against the correct fields.
TC-WF-03	Flag	Records failing one or more rules receive exception status.
TC-WF-04	Classify	Failures receive reason information.
TC-WF-05	Output	Exception output is generated for targeted review.
TC-WF-06	Reconcile	Processed, exception and validation-failure populations reconcile.
TC-WF-07	Report	Governed results are available for Power BI reporting.

8. Power BI Validation Tests

Test ID	Measure / Behaviour	Expected Result
TC-PBI-01	Records Processed	Dashboard total matches the governed processed-record population.
TC-PBI-02	Exception Records	Dashboard uses distinct affected records, not rule-failure rows.
TC-PBI-03	Exception Rate	Dashboard rate recalculates from governed numerator and denominator.
TC-PBI-04	Reason analysis	Reason or rule analysis reconciles to rule-level validation output where available.
TC-PBI-05	Filter context	Record-level measures remain valid under supported report filters.
TC-PBI-06	Turnaround outcome	Dashboard uses 6 hours before and 2.5 hours after per file when showing confirmed turnaround context.
TC-PBI-07	Confidentiality	Public portfolio views contain no PHI or confidential client values.

9. Batch-Scale Test

Attribute	Test Specification
Reference scale	Approximately 250,000 records per batch
Objective	Confirm the portfolio workflow can execute its validation logic against the confirmed engagement-scale population.
Validation focus	Completion of rule execution, exception output generation and reconciliation.
Pass condition	The governed population completes processing and validation outputs can be reconciled.
Boundary	No specific processing-time SLA or infrastructure benchmark is claimed because historical technical timing is not retained.

10. Defect Handling

Stage	Control
Identify	Record the failed test condition and affected requirement or rule.
Classify	Identify whether the issue relates to data, rule logic, exception handling, grain, reconciliation or reporting.
Correct	Update the applicable portfolio implementation logic.
Retest	Repeat the failed test and related regression checks.
Close	Close only after expected results reconcile.
Historical boundary	No historical defect count or severity distribution is claimed.

11. Entry & Exit Criteria

Type	Criteria
Entry	Applicable requirement, rule and test condition are defined.
Entry	Required test data or controlled examples are available.
Entry	Expected pass/fail behaviour is documented.
Exit	Implemented validation checks produce expected results.
Exit	Exception outputs reconcile to the processed population.

Exit	Record-level Exception Rate is not inflated by multi-failure records.
Exit	Power BI measures reconcile to governed validation outputs.
Exit	Known portfolio test issues are resolved or explicitly documented.

12. Requirements Traceability

Test Area	Requirements / Rules	Supporting Artifact
Input and population	BR-01, BR-02	BRD; Automated Validation Workflow
Validation rules	BR-03 to BR-09; DQ-01 to DQ-08	Data Quality Rule Catalogue
Exception handling	BR-10 to BR-13	Exception Analysis
Exception Rate	BR-14; RULE-01	KPI Catalogue
Traceability	BR-17	Data Dictionary
Reporting	BR-15, BR-18	Power BI Requirements; Data Model & DAX Measures
Turnaround outcome	Confirmed project outcome	Findings & Recommendations

13. Test Evidence Boundary

Portfolio integrity: This document demonstrates how the confirmed healthcare validation solution can be tested professionally. It does not fabricate historical pass counts, defect logs, UAT approvals, clinical test data or production performance results.